Q.25 What is Laplace transform? Define transfer function.

Q.26 Write down the advantages and applications of open loop system.

Q.27 Define manually controlled closed loop system. Draw the block diagram of it.

Q.28 Write any five applications of tachometer.

Q.29 What is potentiometer? Explain its working.

Q.30 Define stability. Explain RH Criteria.

Q.31 Apply the RH criteria and check whether the given system is stable or not.

$s^4+2s^3+3s^2+5s+1=0$

Q.32 Define the following

a) Linear system b) Bode plot

c) Time constant d) Laplace transforms

e) Variable reluctance type stepper motor

Q.33 What are the applications of servo motor?

Q.34 Write the differences between signal flow graph and block diagram reduction method to find transfer function.

Q.35 Explain the working of variable reluctance stepper motor.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

Q.36 What is servo motor? What is the working principle of it? Explain all types of servo motor.

Q.37 Explain under damped, over damped and critically damped system with example. What is delay time and settling time of a system? Write all the standard test signals in control system.

Q.38 What is first order system? Find the time response of first order system when subjected to step input.

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181531/121531/031531/073641

3rd Sem / IC, EI

Sub : Basics of Control System / Const. Sys.

Time : 3Hrs. M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

Q.1 Choose the incorrect statement for the open loop control system?

a) It does not have feedback.

b) It is complex in design

c) It is highly sensitive towards the change in forward path parameter

d) Design of such circuit is usually inexpensive

Q.2 Block diagram reduction technique is used to find the _______ of the system?

a) Transfer function of the system

b) Frequency response of the system

c) Gain margin and phase margin of the system

d) Time response of the system

Q.3 In block diagram reduction technique, if two blocks are in parallel than the overall gain of the system is

a) Product of individual gain of both the blocks

b) Division of individual gain of both the blocks

c) Sum of individual gain of both the blocks

d) Exponential of product of individual gain of both the blocks

Q.4 If there is no sign change in first column element of Routh array then

a) Stable
b) Marginally stable
c) Unstable
d) May be either unstable or marginally stable

Q.5 Choose the correct statement about the impulsesignal

a) The magnitude is 1 for time less than 0 and zero otherwise
b) The area is 1 for time greater than 0 and zero otherwise
c) The magnitude is 1 for time equal to 0 and zero otherwise
d) The magnitude does not defined at time equal to 0

Q.6 From the Root locus, we can find

a) Stability of the system
b) Transfer function of the system
c) Frequency response of the system
d) All of the above

Q.7 The system is said to be linear if

a) If follows the principle of superposition and homogeneity
b) It is time invariant
c) It does not follow the principle of superposition and homogeneity
d) None of the above

Q.8 The synchro-pair transforms

a) The electric signal into angular speed
b) The linear speed into electric signal
c) Works as electromechanical switch
d) The angular position of shaft into electrical signal

Q.9 Which of the following is correct about the working principle of potentiometer?

a) Principle of inductance change
b) Principle of resistance change
c) Principle of capacitance change
d) Principle of electric flux change

Q.10 The _____ is the time taken by the system response to reach to the final value.

a) Rise time b) Peak time
c) Setting time d) Delay time

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

Q.11 Open loop system does not have feedback. (True/False)
Q.12 Laplace transforms of the unit step function is _____.
Q.13 Define the settling time.
Q.14 Define servo motor.
Q.15 $s^3 - 3s^2 + 5s + 2 = 0$ is a stable system. (True/False)
Q.16 Potentiometers is an active device. (True/False)
Q.17 If the denominator of a system transfer function become 0 then the value of s is known a _____ of the system. (Pole / Zero)
Q.18 Define stability.
Q.19 Define tachometer.
Q.20 Expand CLTF.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

Q.21 Write advantages and disadvantages of closed loop system.
Q.22 What are the linear and nonlinear systems?
Q.23 Explain servo mechanism.
Q.24 Define rise time, delay time, peak time and maximum overshoot with graph.